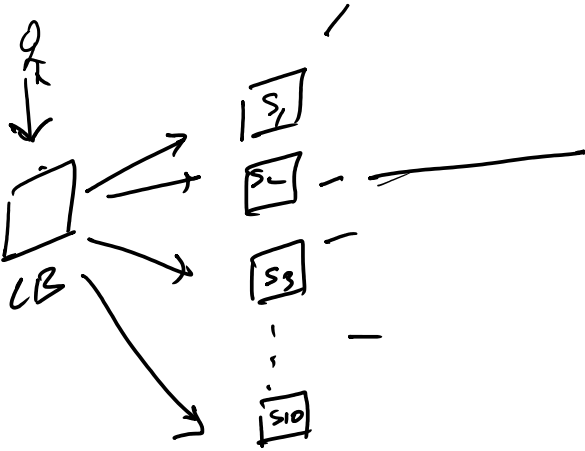


Load Balancer:-



CI/CD Workflows with Approval Gate:-

Control
Auditable Deployment } - enterprise, regulated or
multi-team environment.

Code → Build → Test → Staging Deploy → Approval gate
→ Prod Deploy → Monitor

Detailed CI/CD stages with Approval gate

- ① Source Control → Developer push code to git - Git, Gitlab, BitBucket
- ② Build → Compile code, resolve discrepancy - Jenkins, Maven, Gradle
- ③ Unit & static Test → Run Test, Code Quality, SAST - JUnit, SonarQube
- ④ Package → Create deployable Artifact or Container - Docker, jar, Helm charts
- ⑤ Deploy to staging - Deploy to Non-prod. - Kubernetes, ECS, Azure App Service.

- Deploy to staging - Deploy to main - App server.
- ⑤ Deploy to staging -
 - ⑥ Approval gate (Pre Production) - Manual or Automated gate
b/f production - GitHub PR review, ServiceNow, Jira
 - ⑦ Deploy to Production - ArgoCD, GitOps -
 - ⑧ Monitor & Rollback - Tracing, logs, metrics & error - Dynatrace, Splunk, Prometheus.

Where to place Approval Gate:

- ① Pull req. review - Code push to main - GitHub, GitLab
- ② Pre-prod Manual Gate - After staging tests - DevOps, GitLab
- ③ Change management Gate - B/f production deployment - ServiceNow, Jira.
- ④ Security Scan Gate - threshold exceeded, - SonarQube, Checkmarx.

Example:-

Jenkins + GitHub + ServiceNow Approval gate:-

- ① pipeline trigger to push change to main branch
- ② Jenkins builds code & runs tests.
- ③ After staging deploy:-
- Jenkins send req. to ServiceNow for approval.
- wait for approval.

- jenkins will wait for api
- One Approved Jenkins deploy to production.